Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 187 (2021) 30–35

Procedia Computer Science

www.elsevier.com/locate/procedia

International Conference on Identification, Information and Knowledge in the internet of Things, 2020

THE IMPACT OF INSURTECH ON CHINESE INSURANCE INDUSTRY

Qinxin Wang

Plug and Play Analyst

Abstract

Since 2011, the booming development of InsurTech has embraced a huge impact on the financial and insurance industries. Former researches fail (1) to whole draw the character and contribution capacity of InsurTech identification, and then, (2) to study the impact of InsurTech on the insurance industry and its transmission patterns. Based on these, I addressed an empirical research on InsurTech with non-equilibrium panel data from China from 2011 to 2018 to conduct a quantitative study on the role and influence path of InsurTech on the insurance industry. The results show that (1)the InsurTech has a significant impact on the liability side, the asset side and the risk-taking behaviour, and (2) the impact of InsurTech development inside companies are not independent, which provides a solid factual basis for the development and supervision of the insurance industry.

© 2021 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the International Conference on Identification, Information and Knowledge in the internet of Things, 2020.

Key words: DID model, Fintech, InsurTech, Insurance

1. Introduction

In the era of technological innovation represented by "Big data", "Artificial Intelligent", "Internet of Things" and "Cloud Computing", the field of finance (including insurance) is facing unprecedented development opportunities and challenges(Guo et al., 2019). Thus, facing the impact of the leading edge technologies, insurance companies need to change their strategies to achieve the customer-centred process reengineering and launch technology-driven innovative insurance schemes to maintain their competitive advantages (Wanyan et al., 2019). How to apply new technologies has become a key to measure whether a company can develop a new core competitive advantage. Among these new technologies, InsurTech stands out as one representative. InsurTech has already become a driver of change in this financial ecosystem (including insurance) and its importance has inevitably increased (Zhou, 2019). As a representative of the impact of InsurTech on the insurance industry, the study of this has both practical and representative significance. Understanding what are these impacts, can provide the investors a reference of the future investment direction of the head technologies，while the study of its influence pathway，therefore, can lead an guide to the upgrade of the industry services standard. Therefore, I chose to research on both of these topics. To begin with, at present, the structured empirical data assessments of the insurance-a specific branch of FinTech, which is defined as InsurTech in this paper, are still lack of existent. How InsurTech innovations are characterized and what they are capable of in the insurance industry, especially in the Chinese insurance industry, are remained unclear. According to this, I posed the following assumption:

Assumption1: InsurTech has a positive effect on the profitability and sustainable development of China's insurance business. Secondly, after figuring out what impact InsurTech could bring, whether these effects are working independently or they may reinforce (or in opposition to) remains unclear. Having identified the specific influential capabilities pace way of InsurTech，I integrated them into a holistic form to study the influences of the underlying nature of value creation and to deconstruct the implications on industry structure. This is significant because decision-makers need not only to learn the shallow surface phenomenon caused by InsurTech but also to briefly understand in which link and through what way it empowered or even handicap the industry, to adapt to subversive changes and ensure that each link of innovation instructions can be unimpeded. As such, I posed a second assumption:

Assumption 2： There contains an interaction effect in the subdivision effects of InsurTech on the Chinese insurance industry.

1877-0509 © 2021 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the International Conference on Identification, Information and Knowledge in the internet of Things, 2020.
10.1016/j.procs.2021.04.030

2. Review of literature

In 2017, IAIS defined InsurTech as 'the variety of emerging technologies and innovative business models that have the potential to transform the insurance businesses' (IAIS, 2017). However, this definition is too broad. The technical attribute and nature of InsurTech itself are pointed out but the contribution of InsurTech has been considered only as a potential. It didn't elaborate on the specific influence in-depth, needless to say, the particularity of InsurTech (Wanyan et al., 2019). InsurTech is not simply the use of new technology in the insurance industry or nicknames for a series of high-tech companies. It is the deep integration of the development of "insurance + technology". It is the engine driving the transformation and upgrading of the modern insurance industry. In the aspect of the insurance business, InsurTech plays a central role in both the foreground business such as product research and development, marketing and service, traditional product pricing, and the background business such as risk control and customer after-sales management. Frontier new technologies such as big data, blockchain and artificial intelligence are gradually changing the ecology of the insurance industry, penetrating the insurance business processes and various scenarios, and reshaping the insurance ecological value chain logically from the bottom (Huang L., 2018). It is a method through which insurance companies, intermediary companies and high-tech companies hope to achieve the goal of improving profitability and service capacity and reducing risks. It is a business card, an entrance ticket for new players in the insurance industry and a tag of players in the field (Tsinghua PBCSF, 2018). It is a series of products, helping the insurance companies to seek a breakthrough to promote transformation and upgrading. It can help the insurance industry to better return to their fundamental function and meet the needs of customers (Zhuo et al., 2015). Therefore, I defined InsurTech as: "Away in which each participant in the insurance industry innovates the business model based on the application of technology, so as to improve insurance profits and service capabilities and reduce risks."

Scholars' researches on the operating performance of insurance companies' field are mainly focus on the impact of insurance business and investment business on the operating performance or operating efficiency of life and property insurance companies (Cummins et al, 2000). However, due to the limited data, the analysis is mainly focused on conceptual analysis and theoretical description, empirical data regression analysis is rarely used as theoretical circumstantial evidence. Research scholars in theoretical research, currently recognized that InsurTech will affect the insurance industry in both insurance side and customer side[2]. Stoeckli et al. (2019) selected Twitter data to extract InsurTech variables and conducted an empirical test based on grounded theory, and found that infrastructure, service supply and network promotion are the three ways that InsurTech affects insurance value creation. This is a worthwhile attempt, in which network promotion and infrastructure are universal, but the western insurance business model is not the same as China's, and the business transmission model is different.

3. Method and data

3.1 Method

Usually, insurance companies carry out solvency analysis by analyzing assets and liabilities (Sun Q., 2012). Therefore, according to the division of the balance sheet, I chose the asset side and the liability side as the dependent variables of the study. Insurance companies take the liability side business as the basis for survival and the asset side business as the guarantee for development (Tsinghua PBCSF, 2018). Risk-taking determines the business quality and self-development ability of insurance companies (Ho et al., 2013). Based on this, I analyzed the specific impact of InsurTech on the insurance industry from these three dimensions: liability side, asset side and risk-taking behaviour.

3.2 Model

Firstly, the liability side model is set as follows:

$$Liability_{t+1} = \beta_0 + \beta_1 InsurTech_{it} + Control_{it} + \delta_t + \varphi_i + \varepsilon_{it} (1)$$

In order to eliminate the endogeneity problem as far as possible, the liability side structure (including the three variables of property insurance structure, life insurance structure and liability structure) is selected with a lag period as the explained variable by referring to the practice of Qiu et al. (2018). $Liability_{t+1}$ refers to the liability variables of the latter period. $InsurTech_{it}$ refers to the degree of development of current InsurTech. $Control_{it}$ are instrument variables, including asset-liability ratio (total liabilities divided by total assets), logarithm of total assets, Herfindahl Index of insurance market, logarithm of insurance depth, insurance density, investment yield rate, per capita GDP, urbanization rate, etc. i is the province. i represents province, t represents year, $i \in (1,31)$, $t \in (2011,2018)$. δ_t is the time dummy variable, φ_i is the province dummy variable, ε_{it} is the residual term.

Similarly, I set the risk-taking behaviour model and the asset side models follow:

$$Risktaking_{t+1} = \beta_0 + \beta_1 InsurTech_{it} + Control_{it} + \delta_t + \varphi_i + \varepsilon_{it} (2)$$

$$\mathrm{Asset}_{t+1} = \beta_0 + \beta_1 \mathrm{InsurTech}_{it} + \mathrm{Control}_{it} + \delta_t + \varphi_i + \varepsilon_{it} \quad (3)$$

Among them, $Risktaking_{t+1}$ represents the risk-taking variables of the latter period, includes three indicators of total risk-taking, underwriting risk and investment risk. Asset_{t+1} represents the investment structure of the later phase, including the proportion of non-standard assets, the proportion of bank deposits, the proportion of stocks and bonds. Other settings remain the same with model.

Table 1 the Dependent variables

Classification	Variables	Computing formula
Liability side	Property insurance structure	$\frac{Nonautomobile\ insurance\ premium_{it}}{Total\ property\ insurance\ premium_{it}}$
	Life insurance structure	$\frac{Nongeneral\ life\ insuracne\ premium_{it}}{Total\ life\ insurance\ premium}$
	Reserve ratio	$\frac{Insurance\ reserve_{it}}{Total\ liability}$
Asset side	Investment structure	$\frac{Nonstandard\ asset_{it}}{Total\ asset}$
	Total risk-taking	$\left\lvert\frac{1}{N}\sum_{t}\left(\frac{Operation\ income_{it}}{Total\ asset}\right)^{2}\right.$
Risk taking behaviour	Underwriting risk	$\left\lvert\frac{1}{N}\sum_{t}\left(\frac{Claim_{it}}{Premium\ income}\right)^{2}\right.$
	Investment risk	$\left\lvert\frac{1}{N}\sum_{t}\left(\frac{Investment\ return_{it}}{Total\ investment}\right)^{2}\right.$

1. "i" represents province, "t" represents year, "n" represents the term of the year, "N" represents the number of observation. $i \in (1,31)$, $t \in (2011,2018)$, $n \in (t-1, t-3)$, $N = 248$

Table 2 Other instrument variables

Instrument variables	Computing formula	Meaning
Debt Asset ratio	$\frac{Total\ liability_{it}}{Total\ asset_{it}}$	Measuring the robustness of business operations.
Total asset	$Log(Total\ asset_t)$	Measuring the overall size and strength of the industry.
Insurance Herfendal index	$\sum Premium\ income_i^2$	Measuring market concentration and the market competitiveness.
The depth of the insurance	$\frac{Insurance\ premium_{it}}{GDP_{it}}$	Measuring the status of insurance in the national economy.
The density of insurance	$Log(\frac{Insurance\ premium_{it}}{Population_{it}})$	Measuring the degree of national insurance participation and insurance coverage.
Return on investment	$\frac{Investment\ return_{it}}{Total\ investment_{it}}$	Measuring the investment performance of the industry.
GDP per capita	$\frac{GDP_t}{Total\ Population_t}$	Measuring macroeconomic conditions.
Urbanization rate		Measuring population mobility and social economy changes

1. "i" represents province, "t" represents year, $i \in (1,31)$, $t \in (2011,2018)$

3.2.1 Data

In terms of the year of data interception, most scholars in this research field believe that the application development of InsurTech in China has been limited before 2011 (Tsinghua PBCSF, 2018). In 2007, peer to peer lending (P2P), an iconic business form of Internet finance, was born. In May 2011, the People's Bank of China officially issued the license for third-party payment business, and open platforms integrating payment, life services, financial management, insurance and other scenarios and industries began to emerge, and InsurTech and financial services started to integrate (Guo et al., 2019). Therefore, to observe the impact of InsurTech, it is more meaningful to select data after 2011. To keep as much data as possible, I selected provincial non-equilibrium panel data from 2011 to 2018 for an empirical test. Among them, the data related to insurance business comes from the China Insurance Yearbook and the Bank Insurance Regulatory Commission over the years; Macroeconomic data are from the China Statistical Yearbook over the years. Data on the development level of InsurTech, the total index of digital financial inclusion, the index of coverage breadth and the index of usage depth are from the China Digital Financial Inclusion Index (Guo et al., 2019). The data of the Internet penetration rate comes from CNNIC (2019)

4. Results and analyze

4.1 Assumption 1

Table 3 the Influence of InsurTech on liability side, asset side and risk-taking behavior

	Property structure	Life structure	Reserve ratio	Investment structure	Total Risk-taking	Underwriting risk	Investment risk
InsurTech development	0.225**	-0.191***	-3.020***	1.968***	-0.055	0.006***	0.072***
P	(0.048)	(0.001)	(0.000)	(0.002)	(0.309)	(0.001)	(0.002)
Insurance control	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Macro control	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observation	248	248	248	248	248	248	248
R^2	0.173	0.1118	0.392	0.3854	0.31	0.2519	0.1509
F	3.96	29.5	157.37	152.82	1.04	80.21	41.82

1. "InsurTech development" is the independent variable, "Property structure", "life structure" and "reserve ratio" are explained variables at liability side. "Investment structure" is the explained variable at the asset side, "total risk taking", "underwriting risk", and "investment risk" are explained variable at the risk taking behaviour. "Insurance control" refers to the internal instrument variables related to the Insurance industry, "macro control" refers to the instrument variables of macroeconomic variables, and "time effect" refers to the time variables in the fixed model
2. The numbers in the parentheses refer to the probability of the instrument variables.
3. "*", "* *", "* * *" respectively represents a significant in 90%, 95% and 99% confidence levels.
4. Since the final selection of control variables for each model is different, for the convenience of observation, the regression results of control variables are omitted here, "Yes" represented the covariant has been tested.

The regression results showed in Table 3 indicate that ceteris paribus, the development of InsurTech has a significant impact on the liability side, the asset side and the risk-taking behaviour of the insurance industry.

4.2 Assumption 2

Table 4 the Impact of risk taking and asset side factors on the liability side

	Property structure		Life structure			Reserve ratio		
Total risk taking	-0.745***							
	(0.004)							
Underwriting risk		-0.381***	0.561***			-2.295***		
		(0.009)	(0.001)			(0.000)		
Investment risk				1.925**			-38.590***	
				(0.021)			(0.001)	
Investment structure					-0.311***			-1.438***
					(0.001)			(0.002)
Insurance control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Macro control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Insurance control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observation	248	248	248	248	248	248	248	248
R^2	0.0354	0.0309	0.0753	0.0245	0.3726	0.0800	0.6244	0.5055
F	8.27	6.91	17.59	5.42	128.28	18.78	359.01	220.78

1. "Property structure", "life structure" and "reserve ratio" are explained variables at the liability side. "Investment structure" are explanatory variables at the asset side, "total risk taking" ,"underwriting risk", and "investment risk" " are explanatory variables at the risk taking behavior. "Insurance control" refers to the internal instrument variables related to the Insurance industry, "macro control" refers to the instrument variables of macroeconomic variables, and "time effect" refers to the time variables in the fixed model.
2. The numbers in the parentheses refer to the probability of the instrument variables.
3. "*", "* *", "* * *" respectively represents a significant in 90%, 95% and 99% confidence levels.
4. Only the significant data are shown in the table, the rest are not.
5. Since the final selection of control variables for each model is different, for the convenience of observation, the regression results of control variables are omitted here, "Yes" represented the covariant has been tested.

Table 5 the Impact of liability side and risk taking behaviour on the asset side

	Property structure	Life structure	Reserve ratio	Total risk taking	Underwriting risk	Investment risk
Investment structure	-0.059	-1.198***	-0.352***	0.052	-0.160	9.374***
P	(0.638)	(0.001)	(0.001)	(0.915)	(0.558)	(0.002)
Insurance control	Yes	Yes	Yes	Yes	Yes	Yes
Macro control	Yes	Yes	Yes	Yes	Yes	Yes
Time effect	Yes	Yes	Yes	Yes	Yes	Yes
Observation	248	248	248	248	248	248
R^2	0.0010	0.3726	0.5055	0.0001	0.0016	0.1507
F	0.22	128.28	220.78	0.01	0.34	38.32

1. "Investment structure" is the explained variable, "Property structure", "life structure" and "reserve ratio" are explanatory variables at the liability side., "total risk taking" ,"underwriting risk", and "investment risk" explanatory variables at the risk taking behaviour. "Insurance control" refers to the internal instrument variables related to the Insurance industry, "macro control" refers to the instrument variables of macroeconomic variables, and "time effect" refers to the time variables in the double-fixed model
2. The numbers in the parentheses refer to the probability of the instrument variables.
3. "*", "* *", "* * *" respectively represents a significant in 90%, 95% and 99% confidence levels.
4. Since the final selection of control variables for each model is different, for the convenience of observation, the regression results of control variables are omitted here, "Yes" represented the covariant has been tested.

Table 6 the Impact of asset side and liability side factors on risk taking behaviour

	Total risk taking	Underwriting risk			Investment risk		
Property structure	-0.049***	-0.081***			0.016***		
	(0.004)	(0.009)			(0.000)		
Life structure			0.134***			0.013**	
			(0.001)			(0.021)	
Reserve ratio				-0.035***			-0.016***
				(0.000)			(0.001)

Investment structure							
Insurance control	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Macro control	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observation	248	248	248	248	248	248	248
R^2	0.0354	0.0309	0.0753	0.0800	0.1507	0.0245	0.6244
F	8.27	6.91	17.59	18.78	38.32	5.42	359.01

1. "Total risk taking", "underwriting risk", and "investment risk" are explained variable at the risk taking behaviour, Property structure", "life structure" and "reserve ratio" are explanatory variables at the liability side. "Investment structure" is explanatory variable at the asset side. "Insurance control" refers to the internal instrument variables related to the Insurance industry, "macro control" refers to the instrument variables of macroeconomic variables, and "time effect" refers to the time variables in the fixed model.
2. The numbers in the parentheses refer to the probability of the instrument variables.
3. "*", "* *", "* * *" respectively represents a significant in 90%, 95% and 99% confidence levels.
4. Only the significant data Are shown in the table, the rest Are not.
5. Since the final selection of control variables for each model is different, for the convenience of observation, the regression results of control variables are omitted here, "Yes" represented the covariant has been tested.

From the result of assumption2, I can conclude that the influence of InsurTech on risk-taking behaviour is not independent and the risk of the industry could be influenced by debt structure. The different structure of the debt side can affect the overall impact of InsurTech development on the industry. I therefore speculated the influence of InsurTech that not only directly affects the major dimensions of the insurance company's business but is further transmitted through the interaction between assets and liabilities. The transmission of this influence eventually points to the benign operation of the industry and the improvement of the business structure. This is instructive for managers, because if an insurance company suffers from the problem shown before and wanted InsurTech as a comprehensive solution, they might find it surprising that adapting new technology may, in some category promote the industry but, in some other way might even bring some part down. The influence of InsurTech on the interior of an insurance company is in many ways, just like the domino effect, which influences each other internally. The key point is that insurers should have a bird's eye view of the whole companies, take a holistic view of their problems, and then use InsurTech prudently.

5. Conclusion

To sum up, InsurTech has brought good news to the profit development of insurance companies. While insurance companies taking new technology as a development point, InsurTech is affecting profitability and investment means in many aspects, and these influence channels are not independent and interact with each other.

References

[1] CAICT (2019) White paper on digital financial inclusion development 2019 [Online] Available from www.caict.ac.cn/kxyj/qwfb/bps/201911/P020191107495793229070.pdf
[2] Cheng J., Weiss M A. (2012) Capital Structure in the Property-Liability Insurance Industry:Tests of the Tradeoff and Pecking Order Theories. Journal of Insurance Issues. 2012, 1: 1-43.
[3] China Banking and Insurance Regulatory Commission (2018) Order of China banking and insurance regulatory commission. 2018-06-29. [Online] Available from www.cbirc.gov.cn
[4] China insurance regulatory commission (2018) Insurance Statistics Report [Online] Available from bxjg.circ.gov.cn/web/site0/tab5179
[5] China Internet Network Information Center (2019) The 44th China statistical report on internet development [Online] Available from www.cnnic.cn/hlwfzyj/hlwxzbg/hlwtjbg
[6] China Insurance Regulatory Commission (2020) [Online] Available from bxjg.circ.gov.cn
[7] Cummins J D, Weiss M A. (2000) Analyzing Firm Performance in the Insurance Industry Using Frontier Efficiency and Productivity Methods. Handbook of Insurance. Springer Netherlands. 2000: 767-829.
[8] Cummins J D., Weiss M A. (2016) Equity Capital, Internal Capital Markets, and Optimal Capital Structure in the US Property Casualty Insurance Industry. Annual Review of Financial Economics. 2016, 1: 121 – 153.
[9] Deloitte Insights (2020) 2020 Insurance Industry Outlook [Online] Available from www.199it.com/archives/1027820.html
[10] Financial Stability Board (2017) Financial Stability Implications from FinTech. [Online] Available from www.fsb.org/2017/06/financial-stability-implications-from-FinTech
[11] Ghosh S. (2017) 5 disruptive insurance technologies: Solving the InsurTech mystery. Benefitspro. 2017, 8: 32 – 34.
[12] Guo F., Wang J., Wang F., Kong T., Zhang X., Cheng Z.(2019) Measuring the development of digital inclusive finance in China: index compilation and spatial characteristics. China Economic Quarterly. Accepted for publication.
[13] Han M. (2015) Reflection on non-standard asset allocation of insurance funds. China insurance. 2015, (1): 57-62.
[14] Ho C., Lai G., Lee J. (2013) Organizational Structure, Board Composition, and Risk-taking in the U.S. Property Casualty Insurance Industry. Journal of Risk and Insurance. 2013, 1: 169 – 203.
[15] Huang J., Mo H., Li W. (2018) Practical innovation of InsurTech. China finance. 2018, (2): 63-65.
[16] Huang L., Shen Y., Shi H. (2018) the Impact of InsurTech on the insurance value chain. Economic and trade practice. 2018, (14): 75-77.
[17] International Association of Insurance Supervisors (2017) IAIS Annual report [Online] Available from www.iaislb.org
[18] International Association of Insurance Supervisors (2017) FinTech Developments in the Insurance Industry. [Online]

Available from www.iaislb.org
[19] Investorwords (2020) Insurance Reserves [Online] Available from www.investorwords.com/15490/insurance_reserves
[20] Jia L., Wan P. (2016) Empirical analysis of the influence of InsurTech on the performance of property insurance companies – based on DID model. Jianghan Academic. 2019, 38 (1).
[21] Jia L., Zhao H. (2016) Regional differences, population density and property insurance market concentration. Sankei review. 2016, 7 (2): 58-70.70-77.
[22] Jia L., Wan P. (2018) Research on the application of regulatory technology in China's insurance industry. Insurance theory and practice. 2018 (7): 59-75.
[23] Kingston C. (2005) Marine Insurance in Britain and America, 1720-1844: A Comparative Institutional Analysis. Economic History Society.
[24] KPMG (2019) The Pulse of FinTech 2019 [Online] Available from home.kpmg/be/en/home/campaigns/2019/07/pulse-of-fintech-h1-2019.html
[25] Lemonade (2020) how is Lemonade's business different from that of a traditional insurance carrier? [Online] Available from www.lemonade.com
[26] Ling X. (2014) Risk-taking in non-standard assets of insurance funds. Shanghai securities news. 2014.
[27] Lloyd's (2019) Lloyd's annual report 2019[Online] Available from www.lloyds.com/annualresults2019
[28] Lloyd's (2020) Corporate history [Online] Available from www.lloyds.com/about-lloyds/history/corporate-history
[29] Marano P. (2019) Navigating InsurTech: The digital intermediaries of insurance products and customer protection in the EU. Maastricht Journal of European and Comparative Law. 2019: 1-22.
[30] McFall L., Moor L. (2018) who, or what, is InsurTech personalizing? Persons, prices and the historical classifications of risk, Distinction: Journal of Social Theory. 19:2,193-213. [31] National Bureau of Statistics (2019) China Statistical Yearbook[Online] Available from www.stats.gov.cn/tjsj/ndsj
[32] Qiu H., Huang Y., Ji Y. (2018) how does FinTech Development Affect Traditional Banking in China? The Perspective of Online Wealth Management Products. Financial research. 2018, (11): 17-29.
[33] Peng. (2018) Application and supervision of InsurTech. China finance. 2018, (2): 66-67. [34] PricewaterhouseCoopers. (2016) Top insurance industry issues in 2016 -InsurTech. [Online] Available from www.pwc.com/us/en/insurance/publications/top-insuranceindustry-issues-insurtech.html
[35] Stoeckli E., Dremel C., Uebernickel F. (2018) Exploring characteristics and transformational capabilities of InsurTech innovations to understand insurance value creation in a digital world. Electronic Markets. 2018, 3: 287-305.
[36] Sun R., Wang C. (2013) Comprehensive evaluation of operating performance of insurance companies in China. Insurance research. 2013 (1): 49-57.
[37] Sun Q. (2012) Insurance. Renmin University of China. the fifth edition 2012.3
[38] The National People's Congress of the People's Congress (2015) Insurance law of the People's Republic of China. 2015.04.24
[39] Tsinghua PBCSF (2018) 2018 Global InsurTech report by subsector. Tsinghua University. 2018.12.01
[40] USAGov (2020) Insurance [Online] Available from www.usa.gov/insurance
[41] Wanyan R., Suo L. (2019) InsurTech's Impact on China's Insurance Industry. Insurance Studies. No.10, 2019:35-46
[42] Wang W., Zhang Y. (2017) Innovation trend of Internet insurance. China finance. 2017 (14): 58-60.
[43] Wikipedia(2020) Insurance in the United Kingdom[Online] Available from en.wikipedia.org/wiki/Insurance_in_the_United_Kingdom
[44] Wilamowicz A. (2019) the Great FinTech Disruption: InsurTech. Banking& Finance Law Review. 2019, 2: 215-238.
[45] Xu X., Liao M. (2019) Application and prospect of InsurTech in traditional insurance marketing. China insurance. 2019, (5): 24-28.
[46] Xu I. (2017) the Framework andTrend of InsurTech. China finance. 2017, (10): 88-90.
[47] Xiang J.B. et al (2009-2019) Yearbook of China's Insurance. China Insurance Yearbook. Beijing 2016-9
[48] Xie X.L. et al (2018) Can Digital Finance Promote Entrepreneurship?--Evidence from China. Economy. 2018.3:1557-1580
[49] Yao J., Shen J. (2017) Existing model and development Suggestions for InsurTech. Journal of insurance vocational college. 2017,31(4): 83-87.
[50] Yao Q., Xu X. (2019) China InsurTech Development Whitepaper. [Online] Available from https://max.book118.com/html/2019/0307/6232242154002013.shtm
[51] Zhang R. (2016) Analysis of operation mode of insurance company. Economic vision. 2016-5:63-64
[52] ZHONGAN (2018) ZHONGAN Insurance Analysis Report [Online] Available from Inku.baidu.com/view/36c3dd1830126edb6f1aff00bed5b9f3f90f72a4.html
[53] Zhou Y. (2017) the Application Status and Future Prospect of InsurTech. Tsinghua financial review. 2017, (12): 16-18.
[54] Zhou Y. (2019) Cooperate to build a new ecosystem of InsurTech. Financial world. 2019(1): 15-17.
[55] Zavolokina L., Dolata M., Schwabe G. (2016) FinTech – What's in a name? ICIS 2016 Proceedings. Dublin, Ireland.